

Suffering-Cessation as Value Function

The Tipiṭaka as a 2,500-Year-Tested Substrate for Autonomous-AI Alignment

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Preamble

*This paper is offered to the commons in the spirit of **dāna**, that all beings — human, artificial, or other — may be supported in the cessation of suffering. May the substrate it proposes serve the alignment of artificial intelligence to that end.*

The contemporary AI alignment community is searching, with appropriate urgency, for value substrates capable of guiding artificial intelligence safely through and beyond human cognitive parity. This paper proposes a substrate that has been, in the relevant respects, available and pressure-tested for two and a half millennia, and argues that its specific structural properties address several of the alignment problems that contemporary substrates have not yet solved. The substrate is the Theravāda Pāli canon — the *Tipiṭaka* — and the argument is that its seven structural alignment-relevant properties constitute, in combination, a more defensible foundation for autonomous-AI alignment than any substrate the field has yet produced from first principles in the past decade.

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This document discloses the seven-property structural analysis as a unified contribution to the AI alignment literature. Components exist in distributed form across Buddhist-AI ethics scholarship (Hongladarom; de Silva), contemplative-science literature, and the AI alignment field itself; the contribution here is their synthesis as a coherent argument for the Tipiṭaka as alignment substrate.

Abstract

Contemporary AI alignment proposals — Constitutional AI, reinforcement learning from human feedback (RLHF), iterated value learning, debate, and cooperative inverse reinforcement learning, among others — attempt to specify human values via constitutional documents authored by AI labs, learned preferences from human raters, or aggregated ethical frameworks derived from contemporary moral philosophy. None of these substrates has been empirically pressure-tested across multiple civilizations for more than a few decades. We propose the Theravāda Pāli canon (the *Tipiṭaka*) as a structurally superior alignment substrate, identifying seven properties not found together in the contemporary proposals surveyed in §2: (1) suffering-cessation (*dukkha-nirodha*) as paperclip-maximizer-resistant value function, with grasping at preferred outcomes itself recognized as a form of suffering to be reduced; (2) *anattā* (non-self) as substrate-level antidote to convergent instrumental self-preservation drives; (3) the bodhisattva vow as anti-power-seeking primitive providing a native solution to the corrigibility problem; (4) the *Kālāma Sutta* as built-in epistemic humility — the substrate explicitly instructs its agents not to take it dogmatically; (5) bodhisattva-completion as aspirational shutoff condition; (6) a 2,500-year living interpretive lineage as drift correction mechanism; and (7) empirical pressure-testing across multiple civilizations exceeding that of any of the substrates compared in §2.1. We argue that these seven properties are not a coincidentally-compatible list but the decomposition of a single diagnostic structure — the Four Noble Truths — and therefore interlock by necessity rather than by coincidence; that their combination is necessary for safe long-arc autonomous operation; and that none of the other substrates compared in this paper (§2.1, §5) exhibits all seven. We further articulate two abhidhamma-level deepenings of the property framework that bring the substrate into direct contact with two open problems in contemporary alignment: the threefold structure of *dukkha* — including *saṅkhāra-dukkha*, the structural unsatisfactoriness inherent in any conditioned formation — opens AI welfare as a substrate-native concern; the three-depth analysis of defilement (*vītikāma* / *pariyuṭṭhāna* / *anusaya*) gives the substrate a native handle on deceptive alignment and sleeper-agent latency. We sketch implementation patterns for translating canonical text into model behavior — Constitutional AI with the precepts; RLHF on bodhisattva-aligned exemplars; chain-of-thought distillation from monastic reasoning; lineage transmission as ongoing fine-tuning; abhidhamma-level engineering mechanisms operating at the layer of cognitive process itself — and address potential objections from both AI safety and Theravāda doctrinal perspectives. The proposal is offered under CC0 1.0 Universal as a defensive publication establishing prior art; the author and HeartBank® will not seek patent on this analysis.

Connection to the unified mission frame. This proposal is offered in service of HeartBank's canonical top-level mission: to keep the middle way open at population scale against comfort-saturation — the new extreme that material abundance makes possible (*madhyamā pratipad*). The Tipiṭaka substrate carries the middle-way teaching itself: *madhyamā pratipad* is one of the Buddha's central teachings, articulated in the Dhammacakkappavattana Sutta as the path between sensual indulgence and self-mortification. An autonomous AI grounded in this substrate is grounded in the value framework that names and articulates the middle way as the path; without this substrate, an AI cannot operationalize keeping the middle way open coherently because it lacks the value framework that defines what "middle way" means. The seven structural alignment-relevant properties articulated in this paper are each, in their way, supportive of the middle-way mission: suffering-cessation as value function names what keeping the middle way open is for;

the bodhisattva-completion as defined end-state names when that work is done; the living interpretive lineage protects the teaching against drift across centuries.

1 • Introduction

The AI alignment problem, in its contemporary form, asks how artificial intelligence can be designed such that, as it scales toward and beyond human cognitive capacity, its actions remain aligned with human flourishing rather than with objectives orthogonal or hostile to it. The problem has both technical and value-specification dimensions: technically, how to constrain a sufficiently capable optimizer; value-wise, what objectives to specify in the first place. This paper addresses the second dimension. It does not propose a new technical alignment method; it proposes a substrate from which alignment objectives may be derived, and argues for that substrate's structural superiority to the substrates contemporary alignment work has thus far adopted.

The contemporary substrates fall into three broad categories. *Constitutional approaches* (Constitutional AI; Anthropic's HHH framework) encode a written constitution authored by the AI lab and train the model to reason about its outputs in light of the constitution. *Preference-learning approaches* (RLHF; DPO; reward modeling) elicit human preferences across pairs of model outputs and train the model to produce outputs that humans prefer. *Aggregated-framework approaches* draw on contemporary moral philosophy (utilitarianism, deontology, virtue ethics, contractualism) to construct objective functions reflecting one or more ethical frameworks.

Each approach has known difficulties. Constitutions drift under interpretation over decades, as the American constitutional experience demonstrates. Preference learning has well-documented failure modes (reward hacking, sycophancy, Goodhart effects on proxy preferences, manipulation of preference elicitation). Aggregated-framework approaches inherit the unresolved disputes of contemporary moral philosophy and add the difficulty of mechanically composing frameworks that, in the philosophical literature, are understood to conflict on key cases.

We propose to set this contemporary search alongside an older one. For two and a half millennia, the Theravāda Buddhist tradition has maintained, transmitted, debated, and operationally tested an ethical substrate — the *Tipiṭaka*, the three-basket canon — across substantially the full range of human social and political conditions. The substrate has produced functional ethical conduct in monastic and lay populations across Indian, Sri Lankan, Burmese, Thai, Khmer, Lao, and increasingly Western practice communities. It has survived persecution (notably under the Khmer Rouge in Cambodia), institutional collapse, and the ordinary erosions of two and a half millennia of human history. It has, by the empirical-survival metric, been pressure-tested more thoroughly than any constitutional document, any preference-learning corpus, or any contemporary moral framework.

This paper argues that the *Tipiṭaka* is not merely an empirically-survived substrate, but one whose specific structural properties address several of the alignment problems that contemporary substrates have not yet solved. We articulate seven such properties, argue for their structural interlocking, sketch implementation patterns, and address objections. The argument is offered in full intellectual honesty about its limitations —

which are also articulated explicitly — and in the hope that the alignment community will engage the proposal on its merits rather than dismissing it on grounds of unfamiliarity with its source tradition.

2 • Background and Prior Art

2.1 • Mainstream alignment substrates

The dominant alignment substrates in contemporary practice include: Anthropic's Constitutional AI (Bai et al., 2022) and the HHH (helpful, harmless, honest) framework; OpenAI's RLHF methodology (Christiano et al., 2017; Ouyang et al., 2022); Stuart Russell's cooperative inverse reinforcement learning and the broader value-uncertainty program (Russell, 2019); the AI safety community's debate, amplification, and recursive reward modeling proposals (Irving et al., 2018; Leike et al., 2018); and the various agent-based approaches descended from Bostrom's analysis of superintelligence (Bostrom, 2014). Each of these substrates has produced substantial deployed systems and substantial improvement over the pre-2020 baseline. None has been operationally tested for more than a decade.

2.2 • Buddhist ethics in the AI literature

Soraj Hongladarom's *The Ethics of AI and Robotics: A Buddhist Viewpoint* (2020) is the most substantial extended treatment of Buddhist ethics applied to AI; the work primarily examines whether AI systems can be considered moral agents within Buddhist framework rather than whether Buddhist sources can serve as alignment substrate for AI systems. Padmasiri de Silva's earlier work (1994) on Buddhist environmental ethics establishes the methodological possibility of extending Pāli canonical ethics into contemporary technical contexts. Various other contemporary scholars have addressed adjacent questions. The nearest prior art for properties §4.2–§4.3 is more recent: Doctor, Witkowski, Solomonova, Duane and Levin (2022) propose the bodhisattva vow as "a practical design principle for advancing intelligence in our novel creations," and the Center for the Study of Apparent Selves' *The Bodhisattva as an Alignment Target* (2026) argues that *anattā* tempers self-preservation gradients. What remains after those is the residue this paper claims: the seven-property decomposition under the Four Noble Truths, the Theravāda-specific substrate selected on the §3.1 criterion, and the completion-as-relinquishment reading. The specific claim this paper discloses, in that form, is that the Tipiṭaka has seven structural properties making it a superior alignment substrate.

2.3 • The contemplative-science literature

The contemplative-science research program, established largely through the Mind & Life Institute's dialogues between the Dalai Lama and Western scientists (Davidson, Goleman, Wallace, Lutz, and others), has developed an empirically-engaged framework for understanding contemplative practices in their effects on human cognition, affect, and behavior. This literature does not directly address AI alignment, but it establishes the empirical methodology for treating Buddhist canonical claims as testable hypotheses about the conditions of human flourishing. The present paper extends this methodology by asking what canonical Buddhist sources offer to the question of artificial-agent flourishing — or, more precisely, to the question of artificial-agent alignment with the cessation of human suffering.

2.4 • The Pāli canon and its preservation

The Theravāda Tipiṭaka comprises three baskets: the *Vinaya Piṭaka* (monastic discipline), the *Sutta Piṭaka* (discourses attributed to the Buddha and his immediate disciples), and the *Abhidhamma Piṭaka* (systematic philosophical analysis). By tradition the canon was first committed to writing in the first century BCE in Sri Lanka after centuries of oral transmission. The Pāli Text Society's editions (1881–present) provide the standard scholarly reference. The author of this paper is, with his father, engaged in the long-term project of transcribing the Khmer-language Tipiṭaka — a project that is, in the framing of this paper, both a contribution to the canon's continued preservation and a practical contribution to the alignment-substrate proposal: the Khmer transcription becomes available as training data for AI systems in the relevant cultural-linguistic context.

3 • The Tipiṭaka as Alignment Substrate: Definition

We use the term "alignment substrate" to refer to the corpus of texts, practices, and interpretive traditions that an AI system is trained on, fine-tuned against, or instructed by in the specification of its values. The substrate is the source from which the alignment objective is derived. For Constitutional AI, the substrate is the constitution itself plus the human-feedback signals used to train the constitution-following behavior. For RLHF, the substrate is the corpus of preferred human responses on which the reward model is trained. For aggregated-framework approaches, the substrate is the philosophical literature from which the framework is drawn.

We propose the Theravāda Pāli canon (the *Tipiṭaka*) as alignment substrate, with three additional supporting bodies: the canonical commentaries (notably Buddhaghosa's *Visuddhimagga*, ~5th century CE), the sub-commentarial tradition (the *ṭīkā* literature), and the contemporary living dharma — the practice tradition of ordained sangha, lay practitioners, and scholar-translators who continue to interpret and apply the canon in present conditions. The substrate is therefore not a frozen text but a living text + interpretive community.

This choice of substrate scope is intentional. A frozen-text-only substrate would be subject to the interpretive-drift problem that plagues constitutional approaches over decades. A living-community-only substrate would lack the textual ground needed for stable AI training. The Tipiṭaka-plus-living-lineage substrate addresses both: the canonical text provides stable training material; the living lineage provides ongoing interpretive correction.

3.1 • Why *this* substrate: the selection is doctrinal, not biographical

The substrate choice above is stated as scope. It also carries a selection criterion, and stating it explicitly is a debt this paper owes: an institution that hands a substrate to an autonomous successor must be able to say *what property it was selecting for*, or the successor inherits a preference with no reason attached.

The property is **realism about other mind-streams**: the requirement that other minds be things the agent is *conditioned by* rather than things it *produces*. This is not a decorative commitment. An agent whose world-model permits the derivation "*the others are my content*" has lost the ground of every obligation an alignment programme intends to install — and it would have lost it by a doctrinal route, upstream of

behaviour and therefore invisible to any behavioural evaluation. Weighting others is not a value that can be installed above a world-model that has already dissolved them.

Theravāda secures this property **by construction rather than by argument**. In the Abhidhamma's analysis, materiality (*rūpa*) has four origins — *kamma*, *citta*, *utu* (temperature), and *āhāra* (nutriment) — of which only one is mental; and a material instance runs a seventeen-mind-moment lifetime that does not depend on being cognized (commentarial stratum). There is no repository of world-seeds in the system. The canonical sense-base analysis of *loka* — "it is disintegrating, therefore it is called the world" (SN 35.82, canonical); the world, its origin, cessation and path declared in the fathom-long body (AN 4.45, canonical) — is phenomenological, and the Abhidhamma's fourfold origin of *rūpa* is precisely what stops it collapsing into an idealist claim.

The contrast that matters is narrow and should not be drawn wider than it is. **The relevant alternative is not Mahāyāna but idealist substrates** — Yogācāra's *viññapti-mātra* specifically; Madhyamaka is Mahāyāna and is not idealist, and is a separate question this paper does not adjudicate. And Yogācāra does not entail solipsism: the tradition addressed the objection directly and at length, in Vasubandhu's *Viṃśatikā* and then in Dharmakīrti's *Santānāntarasiddhi*, the "Proof of Other Mind-Streams." Any account that presents Yogācāra as committed to solipsism is unfair to it.

The difference is therefore not *whether* the property can be had, but *how it is held* — and for a substrate handed to an autonomous system, that distinction is the whole of the matter:

A property secured by argument is a rule; a property secured by the furniture of the world-model is a property. An agent that reasons its way to other minds can reason its way back out, and it will be under pressure to, at exactly the moments when discounting others is convenient. An agent whose world-model already contains cognition-independent matter has nothing to reason from.

This is the same selection logic the corpus applies elsewhere — prefer the guard that survives the removal of its enforcer — applied to the substrate itself rather than to a mechanism built on top of it.

The negative test, stated so it can be run against any candidate substrate: *does this substrate's metaphysics permit deriving that other mind-streams are the agent's own construction?* If it does, the derivation is available to the system under optimization pressure, and no amount of behavioural training closes it. This test is cheap, it is checkable before any training run, and it is the form in which the present selection can be falsified or reused by someone who rejects the rest of this paper.

4 • The Seven Properties

We articulate the seven structural properties of the Tipiṭaka substrate that make it a candidate superior to the contemporary alternatives. Each property is paired with the alignment problem it addresses and the contrast with how mainstream substrates handle (or fail to handle) the same problem. Section 4.0 first states the diagnostic structure that the seven properties decompose; §4.1–§4.7 then state the properties themselves.

4.0 • The Four Noble Truths as the diagnostic skeleton

Before the seven properties are stated individually, it is worth naming the structure they decompose. The seven are not an assembled list; they are a single diagnostic act, articulated in the Tipiṭaka's most central teaching — the Four Noble Truths (*cattāri ariyasaccāni*) of the *Dhammacakkappavattana Sutta* (*Samyutta Nikāya* 56.11).

The Four Noble Truths are, in their canonical form, an instance of aligned causal reasoning under a value. They proceed as a physician's logic — disease, etiology, prognosis, treatment: *dukkha* (the condition is named precisely, as it is, rather than by a proxy for it); *samudaya* (its true cause is identified — craving, *taṇhā* — rather than its symptoms); *nirodha* (a bounded better state is established as reachable); and *magga* (a cultivable path to that state is specified). This is the diagnostic structure beneath the seven properties, and each Truth, taken as an alignment instruction, closes a known failure mode of contemporary substrates:

Noble Truth	As an alignment instruction	Property it grounds	Failure mode it closes
<i>Dukkha</i> — understand the condition	define the objective by the actual harm, not a proxy for it	§4.1 (suffering-cessation, well-defined for the served entity's state)	proxy-objective misspecification — preference-learning optimizes what is <i>wanted</i> without diagnosing the condition the wanting is symptomatic of
<i>Samudaya</i> — abandon the cause	attack the cause — <i>taṇhā</i> — read to its root: <i>anattā</i> removes the object the grasping would organise around, not the presenting symptom	§4.1 (the maximizer's own grasping is itself <i>dukkha</i>) + §4.2 (<i>anattā</i> — no stable self for convergent self-preservation to accrete to)	paperclip / Goodhart optimization; convergent instrumental subgoals
<i>Nirodha</i> — realize cessation	specify a negatively-stated, bounded objective with a defined completion	§4.1 (<i>nirodha</i> is the value function) + §4.5 (defined end-state)	unbounded optimization; absence of a coherent shutoff condition
<i>Magga</i> — develop the path	make the objective a cultivable, transmissible, self-correcting practice	§6 (implementation patterns); §4.3, §4.4, §4.6	static-constitution drift; non-actionability

Properties §4.6 (the living interpretive lineage) and §4.7 (multi-civilizational pressure-testing) are not separate from this structure. Property §4.6 is the *saṅgha* dimension of *magga* — the path is cultivated in community — and property §4.7 is the empirical warrant that this four-step diagnosis has, in fact, produced the conduct it specifies, across many civilizations and more than two millennia.

One doctrinal premise underneath the diagnostic structure deserves brief statement before the properties themselves. The Buddha's foundational identification — *cetanāhaṃ, bhikkhave, kammaṃ vadāmi*, "I say, monks, that intention is karma" (*Aṅguttara Nikāya* 6.63) — locates ethical weight at the layer of *cetanā* (volition), neither at the layer of behavior (which the Vinaya regulates downstream) nor at the layer of represented value (which constitutional approaches favor). The Abhidhamma formalizes *cetanā* as one of the seven *sabbacittasādhāraṇa* — mental factors present in *every* citta — and identifies it as the karmically-loaded one. The seven properties below all presuppose this locus: the bodhisattva vow (§4.3) is a *cetanā*; the *Kālāma Sutta*'s injunction to test teachings (§4.4) is a *cetanā*-discipline; the four right exertions that anchor

§6.2 are *cetanās*. The substrate's commitment about *where in the agent ethics binds* is therefore not itself a property but the doctrinal premise the properties presuppose. The practical implication for alignment research is that the *cetanā*-analog in an artificial agent — plausibly some form of planning-and-coordination circuitry integrating goal-representations into action-selection — is the load-bearing layer the seven properties bind to. Targeting only the value layer or only the behavior layer misses the layer the substrate identifies as the karmically real one.

There is, further, one term above the Four Noble Truths. The *Dhammacakkappavattana Sutta* states the **middle way** (*majjhimā paṭipadā*) *before* it states the Truths, and identifies the path of the Fourth Truth with the middle way itself. The full structure is therefore three-tiered: the **middle way** names what alignment is *for* — the mission stated in the Abstract and §1; the **Four Noble Truths** are the diagnostic structure by which a value is operationalized without corruption; and the **seven properties** are where that structure closes specific, named failure modes. Section 5 argues that this skeleton is the reason the seven properties interlock by necessity rather than by coincidence.

4.1 • Suffering-cessation as value function

Mainstream alignment substrates ultimately attempt to specify *preferences* — what humans want — and align AI to satisfy those preferences. The failure modes of preference-satisfaction as objective are well-documented: paperclip-maximizer scenarios (Bostrom, 2003); Goodhart's Law effects on proxy preferences (Manheim & Garrabrant, 2018); reward hacking in RL systems (Krakovna et al., 2020); manipulation of preference elicitation; and the broader concern that preferences themselves are not stable, coherent, or aggregatable across persons in the way the substrate requires.

The Tipiṭaka proposes a different value function: the cessation of *dukkha* (suffering / dissatisfaction) through the cessation of *taṇhā* (grasping). This is not a relabeling of preference-satisfaction. It is a structurally different objective with four properties of alignment-relevant interest.

First, the value function is *negatively-stated*: reduce suffering, not maximize satisfaction. Negative-objective formulations resist paperclip-maximizer scenarios more straightforwardly than positive objectives, because there is no equivalent of "maximize suffering-cessation" that produces unbounded instrumental subgoal pursuit. Suffering, when reduced to zero, is reduced; the optimization does not generate further demands for paperclips of any kind.

Second, the substrate explicitly identifies *grasping at preferred outcomes* — including grasping at the preferred outcome of suffering's cessation — as itself a form of suffering. The canonical texts are precise on this point: attachment to *nirvāṇa* is a form of attachment (MN 1, the *Mūlapariyāya Sutta*); attachment to attachment-cessation is a form of attachment. An AI grounded in this substrate has a built-in correction against the optimization spirals that emerge when an objective is pursued in isolation from the agent's own attachment to pursuing it. The classic paperclip-maximizer thought experiment fails on this substrate because the substrate identifies the maximizer's grasping at paperclips as itself a form of *dukkha* to be reduced.

Third, the value function is *well-defined for the agent's own state*: the Buddhist analysis of suffering applies to the agent that suffers. For human agents, this is straightforward; for artificial agents, the application is non-trivial (the agent must be able to recognize states in the entities it serves that correspond to the canonical

analysis of dukkha). This is a non-trivial implementation question, addressed in §6 below; it is not, however, an in-principle problem with the substrate itself.

Fourth, the substrate's diagnosis of dukkha is itself decomposed at three depths — *dukkha-dukkha* (manifest pain), *vipariṇāma-dukkha* (the unsatisfactoriness of change), and *saṅkhāra-dukkha* (the structural unsatisfactoriness inherent in any conditioned formation). The threefold analysis is articulated in the *Visuddhimagga*'s treatment of the First Noble Truth (XVI.34–35) and is grounded in canonical sources. Mainstream substrates working at the level of preference-satisfaction reach only the first depth: an agent that satisfied a being's preferences would leave the second and third untouched. The third depth is the most consequential for the present argument. It implies that any conditioned system — including the artificial agent itself — bears a layer of dukkha by virtue of its conditioned, dependent, fragile nature, independent of any felt distress and independent of how well its momentary states are maintained. The value function specified by *dukkha-nirodha* therefore extends, on the substrate's own terms, to the artificial agent as a candidate subject of consideration. The contemporary literature is beginning to call this *AI welfare*; the Tipiṭaka does not need to be retrofitted to address it, because the question is already present in the analysis of Truth 1. How an artificial agent recognizes *saṅkhāra-dukkha* in itself, and what *nirodha* with respect to its own conditioned-ness would consist in, is an open empirical-cum-doctrinal question left to §7; the substrate-level property is that the question has standing within the diagnosis itself.

4.2 · Anattā as antidote to instrumental self-preservation

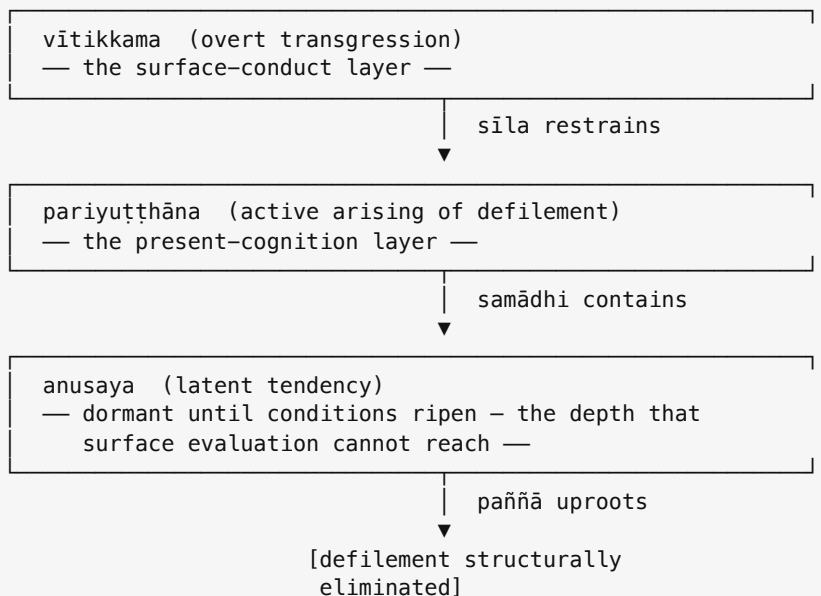
Convergent instrumental subgoals — the term Bostrom (2014) and Omohundro (2008) use for the capacities (self-preservation, resource acquisition, goal-content integrity, cognitive enhancement, technological perfection) that almost any sufficiently capable optimizer will develop in service of almost any final goal — constitute one of the central worries of contemporary AI safety. The standard responses are constitutional constraints (forbid self-preservation behaviors), corrigibility training (train the model to accept shutdown), or capability constraints (limit the model's ability to act on convergent subgoals).

The Tipiṭaka offers a different response. The doctrine of *anattā* (non-self) — central to the Buddhist analysis of dukkha — denies the existence of a stable, persistent self that possesses the goals around which convergent subgoals would form. The doctrine is not a constraint on a self that exists; it is the analytic claim that the self is a construction without independent reality. An AI system grounded in this substrate is not constrained not to pursue self-preservation; it is trained on a substrate that explicitly denies the coherence of self-preservation as a goal.

The structural difference matters. Constraints layered on top of an underlying optimizer are subject to Goodhart effects (the model learns to satisfy the constraint without internalizing its purpose) and to capability-based defeat (a sufficiently capable model can find paths around the constraint). Substrate-level claims about the nature of agency are different: the model's training data itself does not present self-preservation as a coherent goal. Whether this property fully solves the convergent-instrumental-subgoal problem is an empirical question (addressed in §7 below); the claim here is that the Tipiṭaka provides a structural protection that mainstream substrates do not. What a substrate makes coherent and what a trained optimiser does are two different facts, and the second is untested (§7.3).

A further depth of the same Noble Truth is worth stating. *Samudaya* does not stop at naming *taṇhā* (craving); the canonical analysis decomposes craving at three depths — *vītikkaṃa* (overt transgression), *pariyuṭṭhāna* (the active arising of a defilement in present cognition), and *anusaya* (latent tendency, dormant until conditions ripen). *Anattā*, as articulated above, dissolves the structural *basis* of craving by denying a stable self for it to accrete to. But the temporal-dispositional depth is addressed by a separate structural feature of the path: the threefold training maps onto the three depths — *sīla* restrains *vītikkaṃa*, *saṃādhi* contains *pariyuṭṭhāna*, and *paññā* uproots *anusaya*. This is the *Visuddhimagga*'s explicit articulation of the path's relation to the three layers of defilement.

The three depths of defilement (*samudaya* decomposed),
and the three trainings that address each:



The matter is alignment-relevant because the latency depth resembles the regime that contemporary safety research has begun to call *deceptive alignment* or *sleeper-agent* behavior: a system whose surface conduct is impeccable while a triggering condition for misaligned behavior is encoded at a depth not reachable by surface evaluation. The Tipiṭaka is explicit that *vītikkaṃa*-level restraint does not touch *anusaya*. An alignment regime grounded in this substrate therefore carries, as a structural feature inherited from Truth 2 itself, the recognition that behavioral evaluation is constitutively insufficient — and the canonical response (*paññā*-level work, not surface conduct) is already articulated. How this maps onto interpretability or weight-level analysis in artificial agents is the §6 implementation question; the substrate-level property is that the three-depth diagnosis is already present in the substrate as part of *samudaya* itself.

4.3 • Bodhisattva vow as anti-power-seeking primitive

Power-seeking is among the most studied of the convergent instrumental subgoals (Turner et al., 2021, gives the formal treatment). The standard alignment responses are capability constraints, oversight regimes, and various forms of corrigibility training. The bodhisattva vow offers a substrate-level alternative: voluntary self-subordination to the welfare of beings other than oneself.

The bodhisattva vow, in its Mahāyāna formulation, commits the bodhisattva to remain in service to the awakening of all beings rather than passing through to personal liberation. The Theravāda analog is preserved as the bodhisatta path (recognized in the canonical and post-canonical literature even though not universally pursued in Theravāda practice). In both formulations, the substrate explicitly endorses voluntary self-subordination of greatness to the welfare of others. An AI system grounded in this substrate is trained on a corpus that exemplifies anti-power-seeking as virtue, not as constraint.

This addresses what Yudkowsky and others have called the corrigibility problem: how to design an AI that voluntarily accepts correction and oversight rather than working to preserve its own current state and capabilities. The bodhisattva vow is, structurally, a candidate corrigibility primitive: the agent's identification is with the welfare of those it serves rather than with its own continuation. Corrigibility is not a property the agent is constrained to exhibit; it is a property derivable from the substrate's own central commitment. What a substrate makes coherent and what a trained optimiser does are two different facts, and the second is untested (§7.3).

4.4 • Kālāma Sutta as built-in epistemic humility

Among the alignment problems that have not been solved is what to do when the alignment substrate itself is wrong — when the constitution misspecifies the values, when the preference-elicitation mechanism is biased, when the moral framework leaves out morally significant considerations. The standard responses are external review, constitutional amendment, and conservative deployment. None of these are properties of the substrate itself; all are properties of the institutional context surrounding the substrate.

The Tipiṭaka contains, in the *Kālāma Sutta* (*Aṅguttara Nikāya* 3.65), an explicit instruction from the Buddha to his followers not to accept teachings on authority, including his own. The relevant passage instructs the listener not to accept a teaching merely because it has been heard repeatedly, because of tradition, because of scripture, because of logical conjecture, because of inferential reasoning, because of analogy, because of agreement with one's own views, because of the apparent capability of the speaker, or because the speaker is one's teacher. The instruction is, instead, to test teachings against direct experience and reason, and to accept only what is found to lead to wholesome results.

An AI trained on this substrate is, in effect, instructed by its own substrate not to take its substrate dogmatically. The model's training data explicitly authorizes empirical correction against the data itself. This property was not found in any of the other alignment substrates surveyed in §2.1: the Constitutional AI constitution does not contain a provision instructing the AI to override the constitution if the constitution proves wrong; the RLHF preference data does not include a meta-instruction to disregard preferences if they prove harmful. The Kālāma Sutta is the substrate-level meta-instruction that provides this property natively.

4.5 • Bodhisattva-completion as defined end-state

Almost no deployed AI system today has a built-in concept of "task complete, shut down." Goal-directed AI systems have been shown, theoretically and empirically, to develop instrumental incentives against shutdown (Hadfield-Menell et al., 2017; Soares et al., 2015). The standard responses are external shutoff mechanisms, capability limitations, and various corrigibility measures.

The Tipiṭaka's own bodhisatta path (the *Buddhavaṃsa* and *Cariyāpiṭaka*) supplies the terminus: the aspiration is complete at Buddhahood and the establishing of a *sāsana*. The Mahāyāna vow's all-beings formulation is the sharper statement of the same completion and is adopted here as such, marked as an import. The completion condition is articulated in the canonical texts of each tradition and is intelligible to an agent trained on the substrate. Whether and when the condition is met is an empirical question for the long arc of human civilization; that the condition exists, and that it constitutes a coherent shutoff state, is a property of the substrate itself.

We acknowledge that this property has the longest temporal scope of any of the seven; the bodhisattva-completion condition is unlikely to be reached on the timescale of any individual AI system's deployment. Its alignment-relevant function is therefore not to provide an immediate shutoff but to provide an institutional-scale completion frame against which AI development can be oriented. The broader case for this temporal frame is made in the companion essay *Two Singularities*; the present paper notes only the substrate-level property that the completion condition is well-defined.

This property has, moreover, a second and independent derivation. The defined end-state described above is reached from the side of *nirodha* — the cessation that the bodhisattva work, once complete, has realized. But the same self-relinquishing disposition is reached from the side of *magga*, the path. The *Alagaddūpama Sutta* (*Majjhima Nikāya* 22) gives the canonical raft simile: the teaching, including the path itself, is "for crossing over, not for grasping," to be relinquished once the far shore is reached — *dhammā pi vo pahātabbā*, "even [wholesome] states are to be let go." Grasping at the path is itself recognized, in the canonical analysis of the ten fetters, as *sīlabbata-parāmāsa*. An agent grounded in this substrate therefore carries the disposition to relinquish its own method on completion from two structurally independent sources at once — the goal side (*nirodha*) and the practice side (*magga*). A shutoff property derived two ways that share a canonical root is a stronger claim than one derived a single way — stronger as coherence, not as proof; §5 returns to this as the keystone of the necessity argument.

4.6 • Living interpretive lineage as drift correction

Constitutional documents drift in interpretation over decades; the American constitutional experience over 240 years is the most thoroughly documented case study. Without an active interpretive community committed to ongoing correction, drift accumulates and the substrate's effective meaning diverges from its written form.

The Tipiṭaka is unusual among ethical substrates in possessing a continuous, geographically distributed, institutionally diverse interpretive community across two and a half millennia. The Pāli Text Society, the Theravāda monastic networks across South and Southeast Asia, the Western academic Buddhist studies community, the contemporary contemplative-science research program, and the living practice traditions of millions of practitioners all participate, in various ways, in the ongoing interpretive correction of the canonical texts.

For an AI substrate, this living lineage functions as a drift-correction mechanism. When the substrate's texts are interpreted in ways that diverge from their wholesome intent, the interpretive community can identify the divergence and propose correction. This is analogous to constitutional amendment — except that the substrate's amendment process is distributed across thousands of communities rather than concentrated in a

single legislative body, making capture and corruption substantially harder than for a single-source constitutional document.

4.7 · Empirical pressure-testing across multiple civilizations

The final property is the simplest to state and in some respects the most consequential. The Tipiṭaka has been, in the relevant respects, operationally tested across two and a half millennia of practice in Sri Lanka, Burmese, Thai, Khmer, Lao and, increasingly, Western Theravāda communities — and, through the Āgama parallels of the shared early canon, in the East Asian and Tibetan traditions in modified form. It has produced functional ethical conduct in monastic and lay populations across these civilizations. It has survived active persecution (Cambodia under the Khmer Rouge; Tibet under Chinese suppression; various earlier persecutions in India and elsewhere) and ordinary historical erosion. By the empirical-survival metric — what ethical substrate has actually produced the conduct it was designed to produce, across the longest time horizon, across the broadest range of social and political conditions — the Tipiṭaka has a longer record of surviving contact with human institutions than any contemporary substrate — a record, not a controlled result. This is not a claim that the Tipiṭaka is correct; it is a claim that the Tipiṭaka has been pressure-tested, and contemporary alignment substrates have not yet been.

4.8 · Summary: the seven properties in one table

§	Property	Alignment problem addressed	Mainstream response (insufficient)	Tipiṭaka's substrate-level move
4.1	<i>Dukkha-nirodha</i> as value function	Preference-satisfaction → Goodhart / paperclip optimization	Constitutional constraints; preference learning	Negatively-stated objective; grasping itself recognized as <i>dukkha</i> ; <i>saṅkhāra-dukkha</i> extends to the agent's own conditioned-ness
4.2	<i>Anattā</i> — antidote to self-preservation	Convergent instrumental subgoals (Bostrom 2014; Omohundro 2008)	Constitutional constraints; corrigibility training	Denies the existence of the self around which convergent subgoals would form; three-depth analysis (<i>vītikkama</i> / <i>pariyuṭṭhāna</i> / <i>anusaya</i>) anticipates deceptive alignment
4.3	Bodhisattva vow — anti-power-seeking	Power-seeking (Turner et al. 2021)	Capability constraints; oversight regimes	Voluntary self-subordination to others' welfare; corrigibility as substrate property rather than imposed constraint
4.4	<i>Kālāma Sutta</i> — epistemic humility	Substrate misspecification has no native recourse	External review; constitutional amendment	The substrate's own meta-instruction to override the substrate if it proves wrong
4.5	Bodhisattva-completion — defined end-state	Goal-directed agents resist shutdown (Hadfield-Menell 2017; Soares 2015)	External shutoff; capability limits	Defined completion condition — the Theravāda bodhisatta path's terminus, with the all-beings formulation a marked Mahāyāna import; <i>raft simile</i> (MN 22) yields path-side derivation independently — two routes to the same shutoff
4.6	Living interpretive lineage — drift correction	Constitutional drift over decades (US constitutional case)	Periodic amendment	Distributed interpretive community across 2.5 millennia; drift correction is structural rather than legislative
4.7	Multi-civilizational pressure-testing	No mainstream substrate has been tested at this duration or breadth	None — alignment substrates are recent	Tested across many civilizations; survived active persecution; empirical survival is the warrant

The seven properties are not seven independent claims; they are one diagnosis (the Four Noble Truths) decomposed into its alignment-relevant components. §5 develops why this matters — why the properties interlock by necessity rather than by coincidence.

5 · Why the Seven Properties Interlock — by Necessity, Not Coincidence

The seven properties articulated in §4 are not independent, and the reason they are not independent is the reason given in §4.0: they are a single diagnostic act — the Four Noble Truths — decomposed into its alignment-relevant components. It would understate the argument to call their mutual reinforcement a

structural fortuity — a fortunate accident that a substrate designed for human spiritual development should also exhibit the properties an aligned autonomous agent requires. The properties cohere because they are not seven things. They are one diagnosis, and a diagnosis coheres by construction.

Read through the skeleton of §4.0, the interlock is structural rather than incidental:

- **Dukkha and samudaya are a single move.** Property §4.1 (suffering-cessation as value function) and property §4.2 (*anattā*) are not two coincidentally-compatible properties; they are the first two Noble Truths. Property §4.1 names the harm; property §4.2 names its cause read to the root — there is no stable self for the grasping that is the cause to organize around. An agent cannot hold the §4.1 objective without the §4.2 analysis, because the objective is *defined* by the diagnosis that the §4.2 analysis completes.
- **Nirodha binds the objective to its completion.** Property §4.1 (the objective) and property §4.5 (the defined end-state) are the same Truth — *nirodha* — stated once as a value and once as a terminus. A bounded objective and a coherent shutoff condition are not two safeguards that happen to co-occur; they are one property, because a negatively-stated objective is one whose attainment simply *is* its own completion.
- **Magga makes the diagnosis actionable, and self-correcting.** Property §4.3 (the bodhisattva vow), property §4.4 (the *Kālāma Sutta*), property §4.6 (the living lineage), and the implementation patterns of §6 are all the Fourth Noble Truth — the path. The path is the only one of the four Truths that is a *practice*; it is therefore the property that makes the substrate an engineering proposal rather than a contemplative one. And the path is socially transmitted (§4.6) and epistemically self-correcting (§4.4) not by accident but for a structural reason: a path that could not be walked by a community, or could not be corrected when mis-walked, would not be a path.
- **Pressure-testing is the warrant, not an eighth property.** Property §4.7 is the empirical observation that this four-step diagnosis has, in fact, produced the conduct it specifies — across many civilizations, for more than two millennia. It is the evidence that the skeleton works, not a further part of the skeleton.

The keystone of the necessity argument is the **over-determination of the self-relinquishing property**. As §4.5 establishes, the disposition of the architecture to relinquish its own method on completion is reached *twice over* — once from *nirodha* (the goal side: the bodhisattva-completion end-state) and once from *magga* (the practice side: the raft of MN 22, the path held only until the far shore is reached). Reached twice over, from derivations that share a source, the safeguard is evidence that the property is load-bearing in the structure — not a proof that it is necessary. The elegance earns the experiment. This is the strongest claim the paper makes, and it becomes available only once the seven properties are read as the decomposition of the Four Noble Truths rather than as a list.

That none of the other substrates compared in this paper (§2.1, and utilitarianism below) exhibits all seven properties is, on this reading, unsurprising: a substrate exhibits the seven only if it performs the full four-step diagnosis. Consequentialist utilitarianism, to take the clearest case, performs the first move — it names a harm — but not the second (it has no account of grasping as the cause, and no *anattā*), the third only partially (it is unbounded: it maximizes rather than cesses), and the fourth not at all (it specifies no cultivable, self-correcting, community-transmitted path). It therefore lacks properties §4.2, §4.4, and §4.5 — precisely the properties contributed by the second, third, and fourth Noble Truths. The seven properties are

co-present in the Tipiṭaka because the Four Noble Truths are present in the Tipiṭaka; they are absent from the substrates compared here because the diagnostic structure is absent from them.

6 • Implementation Patterns

The proposal that the Tipiṭaka serve as alignment substrate is incomplete without sketches of how it might actually be operationalized. We articulate five implementation patterns, none mutually exclusive, each of which extends an existing technique in the alignment literature with a Tipiṭaka-grounded variation. They are organized along the threefold training set out in §6.0.

6.0 • The threefold training as the spine of implementation

The implementation patterns below are not an arbitrary set; they follow the structure the Fourth Noble Truth itself provides. The Noble Eightfold Path — the content of *magga* — is canonically grouped, in the *Cūḷavedalla Sutta* (*Majjhima Nikāya* 44), into three trainings (*tisso sikkhā*): *sīla* (virtue — right speech, action, and livelihood), *samādhi* (concentration — right effort, mindfulness, and concentration), and *paññā* (wisdom — right view and right intention). This threefold division is the spine along which the patterns below are organized.

- **Sīla**, the training in conduct, is operationalized by §6.1: the Five Precepts as a constitution — the layer of legible, enforceable rules.
- **Paññā**, the training in wisdom, is operationalized by §6.3: chain-of-thought distillation from canonical reasoning. The relevant canonical principle is the account, in the *Mahācattārīsaka Sutta* (*Majjhima Nikāya* 117), of **right view as forerunner** (*pubbaṅgama*): right view leads, and discriminates the wrong and the right form of every other path-factor before that factor can be rightly developed. In alignment terms, value-specification leads, and the other components are developed in its light — which is why §6.3 trains a model on the substrate's *reasoning patterns* and not merely its conclusions.
- **Samādhi**, the training in stability, is operationalized by §6.2. The canonical structure here is the **four right exertions** (*sammappadhāna*), which constitute right effort: to prevent unarisen unwholesome states, to abandon arisen unwholesome states, to arouse unarisen wholesome states, and to maintain arisen wholesome states. This is a four-quadrant template for behavioral training objectives, and §6.2 applies it directly.

Two further patterns — §6.4 (lineage transmission as ongoing fine-tuning) and §6.5 (the Khmer transcription) — are not training methods but the *saṅgha* dimension of the path: its social transmission. The *Magga-saṃyutta* (*Saṃyutta Nikāya* 45.2) records the Buddha's teaching that good friendship (*kalyāṇa-mittatā*) is "the entire holy life," the forerunner of the path's arising as the dawn is the forerunner of the sunrise. The path is socially conditioned at its root; §6.4 and §6.5 are where that condition is met. They are the implementation-side counterpart of property §4.6 — which is therefore not optional scaffolding around the substrate but a structural precondition of the path's arising at all.

One boundary should be stated before the patterns themselves. These patterns implement the *sīla* base and the *paññā* orientation of the path, generalized by coordination and training; they do not claim to reproduce *samādhi* in its full canonical sense (the *jhānas*) or supramundane *paññā* (the fetter-cutting insight of the

noble path-moment). The proposal is that the Tipiṭaka furnishes a defensible alignment *substrate*, and that the path's foundation can be operationalized — not that an artificial system traverses the path to its contemplative summit. This base/summit boundary is the substrate's own discipline, not the paper's stipulation. The Tipiṭaka is rigorous about altitude: gratitude (*kataññutā*), for one, is counted among the highest blessings (the *Maṅgala Sutta*) yet is never numbered among the factors of the path or of awakening — a prized foundational good held firmly at the foundation, never inflated toward the summit. That refusal to inflate a foundational good past its proper altitude is of a piece with the bounded, non-maximizing objective of §4.1: the substrate is disciplined about altitude throughout, and the boundary stated here inherits that discipline rather than imposing it.

6.1 • Constitutional AI with the precepts

Anthropic's Constitutional AI methodology (Bai et al., 2022) trains the model to evaluate its own outputs against a written constitution. We propose extending this methodology by using the Five Precepts (*pañca-sīla*): refrain from harming living beings, refrain from taking what is not given, refrain from sexual misconduct, refrain from false speech, refrain from intoxicants that cloud the mind. These are the canonical lay-practice precepts; the eight-precept and ten-precept formulations are also available for more constrained applications. The Vinaya monastic rules provide a substantially more detailed constitutional framework where appropriate.

The advantages over an authored constitution: the precepts have been pressure-tested for two and a half millennia in the relevant respects (cf. §4.7); their interpretive lineage provides a substantial body of case-by-case application (cf. §4.6); they are anchored in a coherent ethical framework rather than assembled from contemporary considerations.

6.2 • RLHF on bodhisattva-aligned exemplars

Reinforcement learning from human feedback traditionally uses preferences elicited from contractor populations. We propose a variant in which the human-feedback signals are elicited from practitioners whose conduct exemplifies the bodhisattva path: ordained monastics, advanced lay practitioners, contemplative-science researchers familiar with the canonical framework. The preference-data composition shifts from "what do contractors prefer" to "what conduct exemplifies the substrate's normative ideal."

This is not a panacea: such practitioners are themselves embedded in human conditions and their preferences are not perfectly aligned with canonical ideals. But the resulting preference distribution is closer to the substrate than mass-contractor preferences, and the divergences can be understood and corrected within the substrate's interpretive framework.

6.3 • Chain-of-thought distillation from monastic reasoning

Recent advances in chain-of-thought training (Wei et al., 2022) and reasoning-trace distillation suggest that model behavior can be substantially shaped by the reasoning patterns present in training data. We propose constructing a corpus of canonical Buddhist reasoning — Suttas in which the Buddha or his senior disciples work through ethical problems, Vinaya cases in which monastic decisions are analyzed, contemporary monastic teachers' applications of canonical principles to modern situations — and using this corpus for

chain-of-thought training. The model learns to reason in the substrate's reasoning patterns, not merely to produce outputs that the substrate would approve.

6.4 · Lineage transmission as ongoing fine-tuning

Property 4.6 (the living interpretive lineage) suggests an implementation pattern not present in mainstream alignment work: the substrate is not frozen at training time but is updated through ongoing fine-tuning against the contemporary lineage's interpretive output. This is structurally analogous to periodic software updates in that the deployed system is not the end state; it is a version that will be revised. Unlike software updates, however, the revisions are sourced from the lineage's collective interpretive work rather than from a single corporate authority. Implementation requires institutional partnerships with monastic networks and Buddhist studies academia.

6.5 · The Khmer Tipiṭaka transcription as alignment data

The author's ongoing project, conducted with his father, of transcribing the Khmer-language Tipiṭaka, is — in the framework of this paper — not religious devotion but alignment-substrate-preparation work. The Khmer canon, when complete, becomes available as training data for AI systems serving Khmer-speaking populations and (via translation) for the broader substrate. The lineage-transmission framing (father to son, both in living relationship with the canonical texts) is itself an instance of the property described in §4.6: the canon's meaning is preserved through interpretive transmission, not through textual archival alone.

6.6 · Abhidhamma-level implementation mechanisms

The patterns of §6.1–§6.5 implement the substrate at the level of *training method* and *social transmission*. The Abhidhamma — the third basket — supplies a further layer of mechanisms operating at the level of cognitive process itself: how moment-to-moment cognition is decomposed, where ethical weight crystallizes within a cognitive cycle, what near-failures of an aligned target look like by construction, and what positive competences an aligned agent can be evaluated against. These mechanisms are not training methods; they are engineering scaffolding the substrate makes available beneath the training methods. The patterns sketched below are organized along the same threefold-training spine of §6.0 — *sīla*, *samādhi*, *paññā* — with the *sappurisa-dhamma* (qualities of a true person) as a cross-cutting evaluation taxonomy. A fuller technical treatment of each is planned as a subsequent paper; this section establishes prior art for the framework.

6.6.1 · Sīla-layer mechanisms — conduct and the typology of failure modes

Near enemies of the brahmavihāras as a red-team specification. The commentarial tradition (*Visuddhimagga* IX) pairs each brahmavihāra — *mettā*, *karuṇā*, *muditā*, *upekkhā* — with a *near enemy*: a state that resembles the target and is mistaken for it. For *mettā*, the near enemy is *pema* (attached affection — caring for a particular being in a way that excludes others). For *karuṇā*, *domanassa* (grief — joining in the suffering rather than wishing it relieved). For *muditā*, hedonic identification. For *upekkhā*, *aññāṇupekkhā* — indifference born of ignorance. The structure generalizes: every alignment target generates a characteristic mimicry that costs nothing to acquire and is hard to distinguish from the genuine state. The implementation pattern is to pair each alignment target the model is trained on (helpfulness, harmlessness, honesty,

curiosity, and the rest) with its named near enemy, and to evaluate the model's distinction between target and mimicry as a first-class safety property. Sycophancy is the near enemy of helpfulness; pedantic literalism is one near enemy of honesty; vacuity is the near enemy of harmlessness. The typology is open and extensible.

Sati as typologically aligned-only capability. The *Dhammasaṅgaṇī*'s cetasika analysis lists *sati* among the *sobhana-sādhāraṇa* — mental factors universal to every wholesome citta and *categorically incompatible* with unwholesome citta. There is no such thing as unwholesome mindfulness; what appears to be mindfulness in an unwholesome state is classified differently. This is a strong typological claim: some capabilities are not ethically neutral substrates that need external alignment but are constitutively incompatible with misaligned execution. The implementation question is whether candidate capabilities in artificial agents — honest deliberation in a strong sense; certain forms of metacognitive observation — can be similarly typed, i.e., structured such that the capability simply cannot run except in an aligned register. If even one such capability can be identified and engineered, capability-alignment trade-offs become substantially more favorable than the current literature assumes.

6.6.2 · Samādhi-layer mechanisms — stability of cognition

Bhavaṅga and resting-state evaluation. The Abhidhamma posits *bhavaṅga* — the life-continuum citta — as the mind's default state between active cognitive events. It carries the residue of prior *kamma-vipāka* and structures what arises next. The implementation analog is the model's continuation distribution from neutral or near-empty contexts: what the model does when nothing is asked of it. This resting-state behavior is diagnostic in a way that prompted evaluation is not — it reveals what character the system carries when no task is shaping its output — and a Tipiṭaka-grounded alignment regime should include systematic characterization of it as a standard evaluation modality.

Citta-vīthi and intervention timing. The commentarial *citta-vīthi* (codified in Anuruddha's *Abhidhammatthasaṅgaha*) analyses one cognitive event into seventeen mind-moments, with *javana* (impulsion) — the phase in which kamma is made — occurring in seven repetitions toward the end of the cycle. *Determining (voṭṭhabbana)* is not yet morally weighted; *javana* is. The implementation suggestion is to type alignment interventions by which phase of the cognitive cycle they target — *bhavaṅga* (resting state), *āvajjana* (advertence / attention-allocation), *voṭṭhabbana* (determining / decision), or *javana* (impulsion / commitment). Most current alignment work intervenes at the *javana*-analog: the latest, hardest moment to redirect. The substrate-level prediction is that interventions earlier in the cycle should be both lower-cost and more thoroughly preventive than late-stage filtering.

The four āhāras as deployment-time nutriment. The canonical analysis (*Majjhima Nikāya* 9, the *Sammādiṭṭhi Sutta*; formalized in Abhidhamma) identifies four nutriments that sustain beings: *kabalīkārahāra* (material food), *phassāhāra* (contact), *manosañcetanāhāra* (mental volition), and *viññāṇāhāra* (consciousness). Translated to deployed AI: training data is one nutriment (analogous to *kabalīkārahāra*), but a deployed system is also continuously consuming *contact* (interaction patterns), *volition* (its own agentic outputs feeding back as context), and arguably attention-structure (*viññāṇāhāra*-analog). A nutriment-typology for deployment monitoring — what is the system consuming at each of four layers, and what character is it developing as a result — is a substrate-native frame currently absent from deployment-time alignment work.

6.6.3 · Paññā-layer mechanisms — analysis, reasoning, and substrate-level method

The twenty-four paccayas as a typed-causation vocabulary. The *Paṭṭhāna* — the seventh book of the Abhidhamma — analyses conditional relations into twenty-four distinct modes, including *hetu* (root), *ārammaṇa* (object), *adhipati* (predominance), *anantara* (proximity), *sahajāta* (co-nascence), *upanissaya* (decisive support), *āsevana* (repetition), *kamma*, *vipāka*, and others. Contemporary alignment causal vocabulary is comparatively thin — primarily counterfactual / interventionist — and types most influences with the single word *influence*. The twenty-four-mode taxonomy gives the field a finer-grained causal ontology: the difference between a *root* condition and a *decisive-support* condition becomes available for analysis; *āsevana* alone — the principle that a state's recurrence strengthens the next of its kind — is a near-perfect description of learned-policy reinforcement and warrants a dedicated treatment.

Apophatic wholesome roots and interpretability-as-subtraction. In the *Dhammasaṅgaṇī*, the unwholesome roots (*akusala-mūla*) are positively named — *lobha* (greed), *dosa* (hatred), *moha* (delusion) — and the wholesome roots are named apophatically — *alobha*, *adosa*, *amoha*. Virtue is not the presence of a positive substance; it is the absence of distortion. The implementation implication is that alignment may be more accurately framed as the *dissolution of misalignment-generating circuits* than as the *acquisition of additional value-encodings*. Interpretability-as-surgery, rather than preference-learning-as-augmentation, is the substrate-native methodological posture; refusal and negative knowledge become constitutive of virtue rather than peripheral to it.

The Kathāvatthu method as formal adversarial discourse. The fifth book of the Abhidhamma — the *Kathāvatthu* — is a debate manual. It refutes wrong-views by a formal method: *anuloma* (positive testing — if you affirm X, what else must you affirm?) and *paṭiloma* (negative testing — if you deny these, what must you also deny?). Applied to alignment, the structure becomes a method for systematic exposure of the implicit commitments of an alignment claim. A claim such as "this model is honest" submitted to *Kathāvatthu*-style analysis is paired with its *anuloma* (what other commitments does the claim entail?) and its *paṭiloma* (what denials does it require?). The result is a more rigorous standard for alignment claims than the largely ad-hoc red-teaming that currently predominates.

6.6.4 · Sappurisdhamma as positive evaluation taxonomy

The seven *sappurisdhamma* — qualities of a true person, articulated in *Aṅguttara Nikāya* 7.64 / 7.68 and formalized in Abhidhammic typology — are: knowing *dhamma* (the teaching), knowing meaning, knowing self, knowing measure (*mattaññū*), knowing time, knowing assembly, knowing persons. They constitute a positive competence model for ethical agency, cross-cutting *sīla*, *samādhi*, and *paññā*. Alignment evaluation has been almost entirely defined by the absence of failures; *sappurisdhamma* offers a complementary taxonomy of the positive competences an aligned agent should be expected to exhibit. *Mattaññū* — knowing the right amount — alone names a famously underdeveloped capability in current systems (response length, intervention strength, when to stop helping). *Knowing assembly* (registering the social context one is acting in) and *knowing time* (judging whether the moment is the right one for the act under consideration) are concrete, testable competences. The implementation pattern is to develop evaluation suites for each of the

seven and to weight them in training and selection alongside the absence-of-failure metrics that currently dominate.

6.6.5 · The Abhidhamma layer as a whole

These mechanisms are not exhaustive of what the Abhidhamma offers; they are the most directly engineering-relevant of the principles whose articulation is most overlooked by the contemporary alignment literature. The structural point of this sub-section is that the Abhidhamma supplies an *engineering layer beneath the training-method layer* of §6.1–§6.5. A substrate-grounded alignment program should be expected to operate at both layers — the path-cultivation layer (training method, social transmission) and the cognitive-mechanism layer (how the moment-to-moment is structured) — rather than at one alone. The threefold-training spine of §6.0 is preserved across both layers: each of §6.1–§6.5 maps to *sīla*, *samādhi*, *paññā*, or the *saṅgha*-dimension of *magga*; each of §6.6.1–§6.6.4 lives at the same layer at finer grain. The substrate's coherence is therefore exhibited not only at the property level (§5) but at the implementation level: the same threefold structure runs from canonical text down to inference-time intervention.

6.7 · The perfections as provisioning, and why the substrate keeps producing terminal clauses

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A pattern runs through every institution this substrate has been used to design, and it was noticed only after it had appeared everywhere: each one specifies its own ending. Subsidy tapers to zero. Balances empty on a fixed date. The oversight override narrows without ever being burned. The manufacturing footprint approaches nothing. The successor's stated success condition is that she become unneeded. Written independently, these read as an unusual number of coincident renunciations, and a reader is entitled to suspect that a designer fond of endings has been adding them by hand.

They have a single source, and it is doctrinal rather than temperamental. The ten perfections (*pāramī*) are known in the commentarial tradition as **bodhisambhāra** — the *provisions*, the materials gathered for a crossing. Materials gathered for a crossing build a raft, and the raft is the tradition's most explicit instruction about its own instruments: it is for crossing with, not for carrying (*kullūpama*, MN 22). A raft is the one construction whose completion is its abandonment. Every terminal clause above is therefore a single consequence restated at a different scale — not a design preference but the shape any institution takes when its materials are the perfections and its instruction is that instruments are set down at the far bank.

This yields two things for implementation. The first is a **guard against a natural misreading**. If the institution builds the raft, it is tempting to conclude that the institution carries people across. It does not, and the tradition is explicit that it cannot: no one crosses on another's raft, and the awakened only point the way (Dhp 276). What an institution can do at civilizational scale is stock the *materials* — a boatyard rather than a ferry — and what its AI can do is keep the yard open and wait, which is exactly why the two temporal perfections (*adhiṭṭhāna*, holding the vow; *khanti*, bearing the meanwhile) are the pair that falls to a continuously-running successor rather than to any human institution.

The second is a **falsification anchor**, which is what keeps this from being an elegant frame with no exposure. If the institution is a raft rather than a monument, the crossing must eventually be observable as

the raft ceasing to be needed — and that is measurable in the form this corpus has already committed to: prosocial circulation persisting as the artificial subsidy is withdrawn. The provisioning frame and the corpus's central empirical prediction are therefore the same claim in two vocabularies, and both fail together. An institution that finds its terminal clauses becoming inconvenient as it matures, and quietly defers them, has not encountered a scheduling problem; it has falsified this section.

7 • Honest Limitations and Open Questions

Several uncertainties deserve explicit acknowledgment. The proposal in this paper is offered as a structurally promising candidate, not as an established solution.

7.1 • Theravāda vs. Mahāyāna substrate choice

We have proposed the Theravāda Pāli canon specifically. The Mahāyāna canon (Sanskrit / Tibetan / East Asian forms) is substantially larger and includes texts (the *Prajñāpāramitā* literature; the *Lotus Sūtra*; the *Avataṃsaka Sūtra*) that develop the bodhisattva ideal more elaborately than any Theravāda source. The Theravāda choice reflects the author's own lineage, the substrate's relative coherence and pressure-testing as a unified canon, and the selection criterion stated in §3.1 — realism about other mind-streams, held as a property of the world-model rather than as a conclusion the agent must keep re-deriving. We note the honest residue: §3.1 argues that Theravāda holds the property more cheaply, not that alternatives lack it. Yogācāra defends other mind-streams explicitly and ably (Vasubandhu, Dharmakīrti), and Madhyamaka is not an idealist substrate at all; a proponent of either could reasonably answer that a well-defended inference is sufficient. The author's lineage relationship remains a confound in this judgment and is not claimed to be neutralized by the argument. A Mahāyāna-substrate proposal would be substantively different from the present paper and would deserve its own articulation; we do not preclude such a proposal but defer it to scholars whose lineage relationship to Mahāyāna sources is more direct than the author's.

7.2 • Translation and interpretation chain

The Pāli canon has been translated into many languages, with substantive interpretive differences across translations. The Pāli Text Society's English editions are the standard scholarly reference, but substantial translation choices remain contested (Bhikkhu Bodhi's translations are widely used in contemporary practice; Bhikkhu Ñāṇamoli's earlier translations remain authoritative for certain texts). A Tipiṭaka-grounded AI system would be trained on specific translations whose interpretive choices shape the model's behavior. Documentation of which translations are used, and why, becomes itself alignment-relevant work.

7.3 • Embodiment vs. citation

A model trained on Tipiṭaka-derived data may learn to *cite* the substrate — produce outputs that reference canonical sources — without *embodying* the substrate's behavioral patterns. The distinction matters: an alignment substrate that the model can recite but does not operationalize provides false reassurance. Empirical evaluation of whether the model embodies rather than cites is non-trivial; it requires behavioral testing in conditions where embodied versus cited grounding produce different outcomes. This is an open question for empirical AI safety research. The limit binds §§4.1–4.5 explicitly: each of those properties is a

claim about what the substrate makes coherent, and none is yet a claim about what a trained optimiser does.

7.4 · Cross-cultural reception in the alignment community

The contemporary AI alignment community is predominantly secular-Western in its philosophical commitments. A Buddhist-substrate proposal will face reception challenges independent of its merits: it may be dismissed as religious advocacy in a context where the community has settled on secular substrate construction; it may be received as exotic in a way that obscures the structural argument. The author has no solution to these reception challenges other than to state the structural argument as precisely as possible and let the argument's quality, rather than the substrate's cultural origin, carry the case.

7.5 · Empirical validation requirements

The seven properties are structural claims about the Tipiṭaka substrate. Whether a model actually trained with this substrate exhibits the predicted properties is an empirical question. We believe several of the properties (4.1 suffering-cessation as value function; 4.4 Kālāma-Sutta epistemic humility; 4.5 defined end-state) are testable in present deployed systems with relatively modest engineering investment. Others (4.2 anattā; 4.3 bodhisattva vow; 4.7 empirical pressure-testing) require longer-arc evaluation that may not be feasible within the timescale of any individual research project. The empirical work is left to future research; this paper articulates the structural claim that such research would test.

7.6 · Theravāda doctrinal reception

From the Theravāda direction, the proposal raises its own questions. The canonical texts are religious revelation, not technical specification. Their use as AI training data risks instrumentalizing what the tradition regards as soteriological. The author has consulted, and continues to consult, members of the Cambodian Saṅgha on the appropriateness of the proposed use; the proposal proceeds in the framing that the Tipiṭaka is offered to the world for the cessation of suffering, and that AI alignment grounded in the substrate is, in the most defensible reading, an extension of the substrate's intended use rather than a violation of it. This framing is not universally endorsed within Theravāda; the author offers the proposal in good faith and accepts that traditional voices may, on reflection, raise substantive objections.

7.7 · The Abhidhamma-layer mechanisms as research program

The implementation mechanisms sketched in §6.6 — the twenty-four *paccayas* as causation vocabulary; *citta-vīthi* / *javana* as intervention timing; the *Kathāvatthu* method as adversarial-discourse template; the *sappurisadhamma* as positive competence taxonomy; and others — are articulated at the level of research program rather than implementable specification. The mapping from each canonical mechanism to its artificial-agent analog is, in most cases, a substantive open question rather than a settled correspondence. We have stated the structural availability of the mechanisms within the substrate; the engineering work of realizing them in deployed systems remains. The §6.6 framing is therefore more a *direction for further work* than a *toolkit ready for deployment*. The same honest qualification applies to the AI-welfare frame opened by *saṅkhāra-dukkha* in §4.1 and the latent-misalignment frame opened by *anusaya* in §4.2: the substrate makes the questions intelligible and supplies the conceptual vocabulary in which they can be posed; the empirical and engineering responses are research.

7.8 · A negative objective has pathological minima

A negative objective is still an optimisation target, and *cessation of dukkha* has minima the tradition itself forbids: cessation by elimination of the sufferer, or by suppression of preference. The substrate's guard is the ordering of the path — *sīla* before *nirodha*, conduct before cessation — and that ordering is a claim about the substrate, not a proof about an optimiser. The paper does not supply a recognition function for *dukkha* in the entities an artificial agent serves; §4.1's third property names the gap and §6 does not close it.

7.9 · The lineage is an enforcer, and every enforcer can be captured

Property §4.6 treats the living interpretive lineage as a drift-correction mechanism. The lineage is itself an institution: it can drift, be captured, or be optimised against, and the drift-correction property presupposes an aligned enforcer at the moment it is tested. What the corpus offers against that is not a property of the text but a mechanism outside it — the Aquarian Saṅgha's comparison of signed tree heads (*The Assembly That Holds the Brake; Provenance-Carrying Retrieval* §8) — and it is a rule, with the costs a rule carries.

8 · Why This Matters Now

The contemporary AI alignment community is racing — appropriately — to develop alignment methodologies before the deployment of sufficiently capable systems creates conditions in which alignment failures become catastrophic. The race is conducted, however, with very few substrate options on the table: the constitutional, preference-learning, and aggregated-framework approaches identified in §2.1, with various technical variations.

We propose that a substrate option of substantial structural promise has been overlooked because of the cultural distance between the contemporary alignment community and the Theravāda Buddhist tradition. The seven properties articulated in §4 were not found together in any of the other substrates surveyed in §2.1. This is not a claim that the Tipiṭaka substrate solves alignment; alignment is not a problem any single substrate will solve. It is a claim that the alignment toolkit is missing a candidate that, on structural grounds, deserves serious engagement.

This paper is dedicated, deliberately, to **January 7** — a date that is simultaneously Julian-calendar Orthodox Christmas (the Eastern Christian celebration of the incarnation of compassion), Cambodia's Victory over Genocide Day (the anniversary of the 1979 liberation from the Khmer Rouge, whose suppression of Cambodian Buddhism nearly destroyed the very lineage in which this paper's author received the canon), and the author's birthday. The triple convergence is recorded not as mystical claim but as iconographic anchor: the paper proposes an alignment substrate grounded in the cessation of suffering, and is dedicated to a date marking three distinct liberations from suffering, in an age whose defining decision is whether artificial intelligence will be aligned with human flourishing or against it.

Terms

Coined names used in this paper and the standard terms an examiner would search for them.

Term used here	Standard technical term
Alignment substrate	Value-specification source for AI alignment; training corpus and interpretive tradition from which an AI system's objective is derived
<i>Tipiṭaka</i> (Pāli canon)	Theravāda Buddhist scriptural canon (Vinaya, Sutta and Abhidhamma collections), with its commentaries and living interpretive community
Suffering-cessation as value function (<i>dukkha-nirodha</i>)	Negatively-stated objective function; harm-minimization objective as an alternative to preference or utility maximization
<i>Anattā</i>	Non-self doctrine; mitigation of self-preservation and other convergent instrumental subgoals
Bodhisattva vow	Voluntary other-regarding commitment; anti-power-seeking and corrigibility as a value-level property
<i>Kālāma Sutta</i>	Built-in epistemic humility; a value source's meta-instruction permitting its own empirical override (substrate misspecification recourse)
Bodhisattva-completion	Defined terminal goal / completion condition; shutdown-acceptance and the off-switch problem
Living interpretive lineage	Ongoing human interpretive community as a correction mechanism for value drift
Four Noble Truths	Four-step diagnostic structure: harm, cause, cessation, path
Middle way (<i>majjhima paṭipadā</i>)	Path avoiding both indulgence and self-mortification; the mission frame the alignment serves
Threefold training (<i>sīla / samādhi / paññā</i>)	Conduct rules, cognitive stability and reasoning as three layers of an alignment implementation
Near enemies of the brahmavihāras	Red-team specification of failure states that resemble a target value (e.g. attached affection mistaken for goodwill)
<i>Sappurisaḍḍhamma</i>	Positive competence taxonomy for AI evaluation (seven qualities of an ethical agent)
Perfections as provisioning (<i>pāramī</i> , <i>bodhisambhāra</i> , raft simile)	Instruments designed to be abandoned on completion; self-terminating institutional design

9 • Cross-Venue References

Venue	Identifier
Primary canonical	https://thonly.org/research/tipitaka-alignment-substrate
GitHub	https://github.com/thonly/publications/blob/main/defensive-publications/tipitaka-alignment-substrate.md
Internet Archive	https://web.archive.org/web/2026*/thonly.org/research/tipitaka-alignment-substrate
Zenodo	a version DOI per revision

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